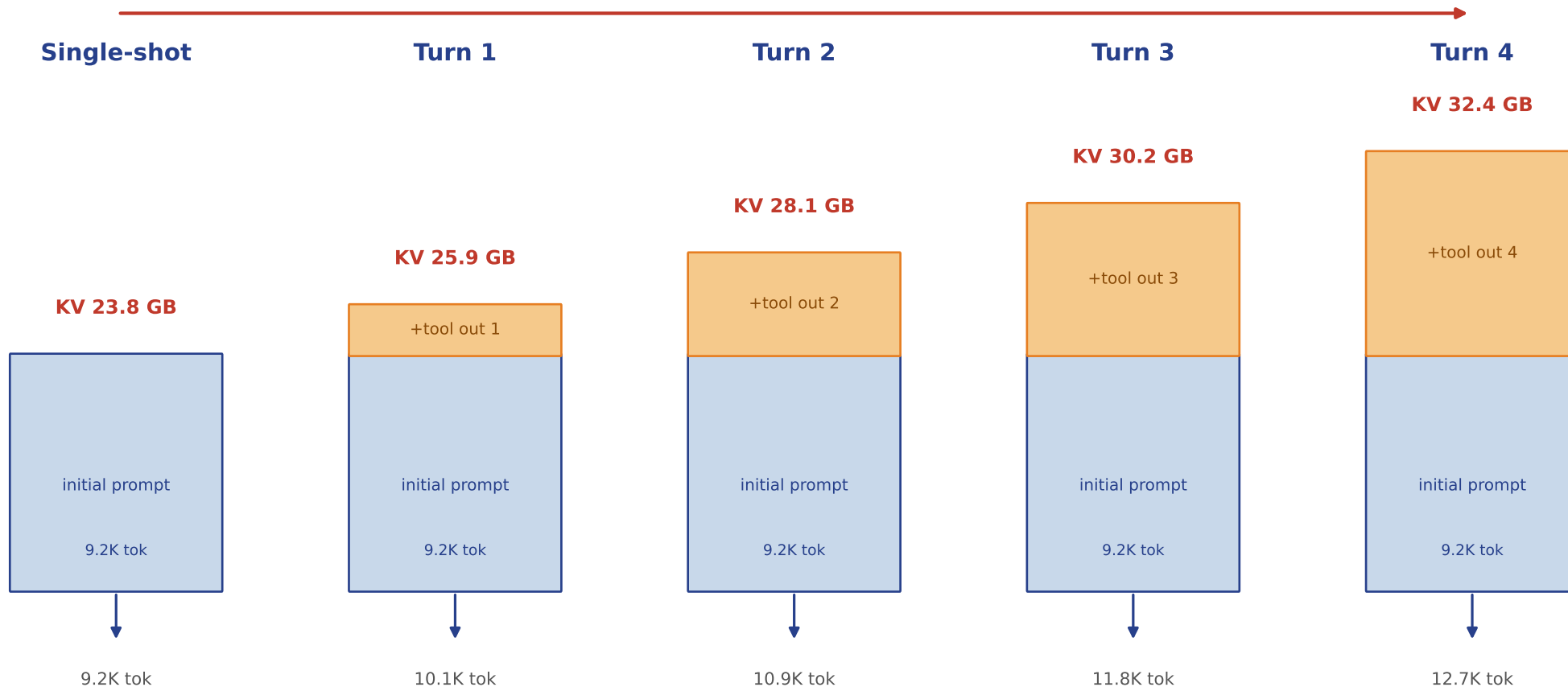


Agentic context accumulation: the KV block grows every turn

each turn appends retrieved context ($\delta=800$ tok) + generated reasoning ($\gamma=65$ tok); tokens persist across all later turns

KV footprint grows ~36% (23.8 → 32.4 GB) — agentic depth is a fleet-sizing problem



$KV = 2 \times n_{\text{layers}} \times d_{\text{hidden}} \times \text{bytes/token} \times \text{total-tokens}$ (2.5 MB/token FP16); total tokens = $l_0 + T \cdot \delta + T \cdot \gamma$. [2° DERIVED, Ch19 §3]